Gemma

#CoP_KIPerWeb

44. Sitzung der offenen Community of Practice

Dr. Andreas Fischer| 12.06.2026

Forschungsinstitut Betriebliche Bildung

Agenda

- Update (allgemein & persönlich)
- Input: „Gemma“
- Diskussion

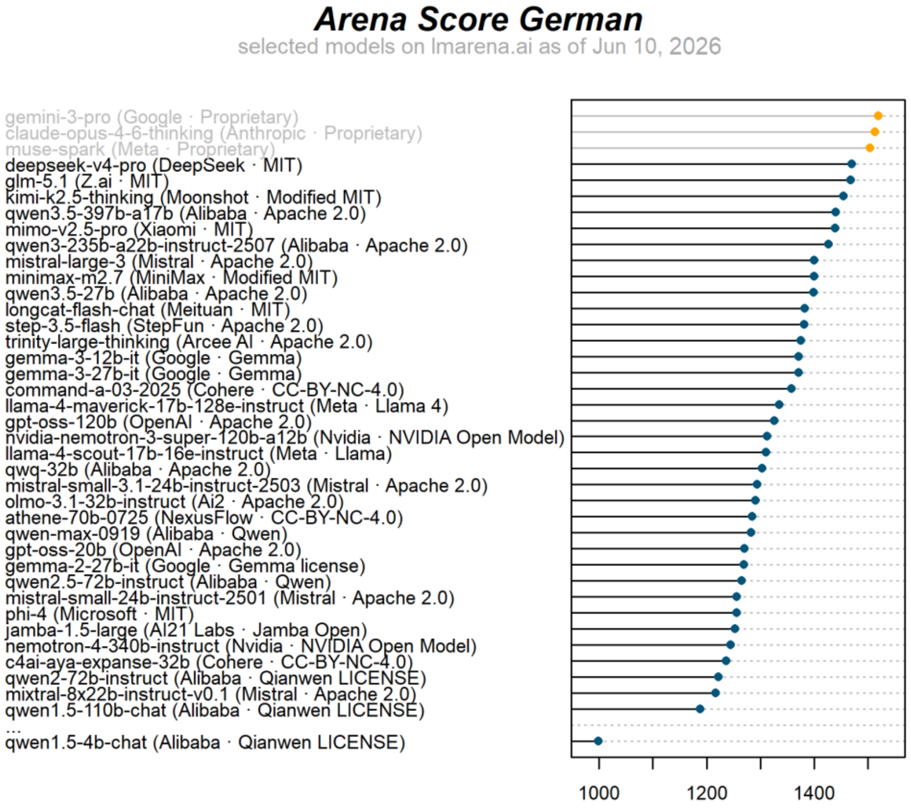
Update (allgemein)

Hintergrund:

- Rechts abgebildet ist auszugweise das aktuelle LMArena-Leaderboard für die Kategorie „German“ mit Blick auf die besten open-weights-Modelle
- je 1 pro Modellfamilie, zzgl. kleine Modelle (<30B) sowie die Top-3 proprietären Modelle

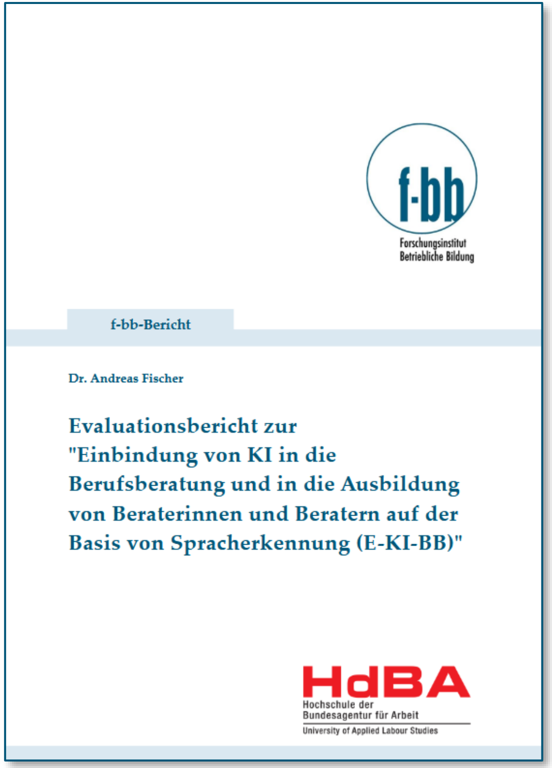
Bemerkenswert:

- Anthropic‘s Claude-opus-4-6-thinking wird an der Spitze abgelöst durch Gemini-3-pro
- Bestes Open-Weights-Modell bleibt Deepseek-v4-pro (1.6T param. total, 49B active)
- Effizientestes Open-Weights Modell bleibt Qwen3.5-27b
    - wobei aktuelle Open-Weights-Modelfamilien wie Qwen3.6 oder die Gemma-4 noch ausstehen
    - „Text (Overall)“ weist Gemma-4-Modelle aus (s.u.)



Update (persönlich)

- Comic „das Kollegium“ mit KI generiert
- Beitrag zu **KI in der beruflichen Orientierung und Beratung** eingereicht (auf Basis unserer Evaluation des E-KI-BB- Projekts, s.Abbildung)
- Beitrag **„Kompetenzprofile und Tätigkeitsanforderungen in der Transformation der Automobilindustrie. Ein KI-basierter Analyseansatz“** akzeptiert zur Publikation in der Zeitschrift für Berufs- und Wirtschaftspädagogik (ZBW)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f-bb-Bericht

Dr. Andreas Fischer

Evaluationsbericht zur "Einbindung von KI in die Berufsberatung und in die Ausbildung von Beraterinnen und Beratern auf der Basis von Spracherkennung (E-KI-BB)"

HdBA
Hochschule der Bundesagentur für Arbeit
University of Applied Labour Studies

Quelle: https://www.f-bb.de/unsere-arbeit/publikationen/evaluationsbericht-zur-einbindung-von-ki-in-die-berufsberatung-und-in-die-ausbildung-von-beraterin/

Fokusthema „Gemma“

Gemma (language model)

7 languages

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From Wikipedia, the free encyclopedia

Gemma is a series of source-available large language models developed by Google DeepMind. It is based on similar technologies as Gemini. The first version was released in February 2024, followed by **Gemma 2** in June 2024, **Gemma 3** in March 2025, and the free and open-source **Gemma 4** in April 2026. Variants of Gemma have also been developed, such as the vision-language model **PaliGemma** and the model **MedGemma** for medical consultation topics.

Gemma

Developer	Google DeepMind
Initial release	February 21, 2024; 2 years ago[1]
Stable release	Gemma 4 / April 2, 2026; 2 months ago[2]
Type	Large language model
License	Gemma 4: Apache 2.0 Gemma 3 and earlier: Source-available (Gemma Terms of Use)[3]
Website	deepmind.google/models/gemma/

History [edit]

In February 2024, Google debuted Gemma, a collection of source-available LLMs that serve as a lightweight version of Gemini. The initial release came in two sizes: neural networks with two and seven billion parameters respectively. Multiple publications viewed this as a response to competitors such as Meta releasing source code for their AI models, and a shift from Google's longstanding practice of keeping its AI source code private.[4][5][6][7]

Gemma 2 was released on June 27, 2024,[8] and Gemma 3 was released on March 12, 2025.[9][10] On April 2, 2026, Google released Gemma 4 under the free and open-source Apache 2.0 license.[2][11]

Quelle: https://en.wikipedia.org/wiki/Gemma_(language_model)

Gemma

Google DeepMind

2024-02-21

Gemma: Open Models Based on Gemini Research and Technology

Gemma Team, Google DeepMind[1]

This work introduces Gemma, a family of lightweight, state-of-the art open models built from the research and technology used to create Gemini models. Gemma models demonstrate strong performance across academic benchmarks for language understanding, reasoning, and safety. We release two sizes of models (2 billion and 7 billion parameters), and provide both pretrained and fine-tuned checkpoints. Gemma outperforms similarly sized open models on 11 out of 18 text-based tasks, and we present comprehensive evaluations of safety and responsibility aspects of the models, alongside a detailed description of model development. We believe the responsible release of LLMs is critical for improving the safety of frontier models, and for enabling the next wave of LLM innovations.

Die erste Gemma-Generation (2B & 7B) erhielt am 5.4. ein Update auf v1.1, sprach allerdings noch kaum Deutsch

Quelle: https://storage.googleapis.com/deepmind-media/gemma/gemma-report.pdf
https://ai.google.dev/gemma/docs/releases?hl=de

Gemma 2

Google DeepMind 2024-06-27

Gemma 2: Improving Open Language Models at a Practical Size

Gemma Team, Google DeepMind[1]

In this work, we introduce Gemma 2, a new addition to the Gemma family of lightweight, state-of-the-art open models, ranging in scale from 2 billion to 27 billion parameters. In this new version, we apply several known technical modifications to the Transformer architecture, such as interleaving local-global attentions (Beltagy et al., 2020a) and group-query attention (Ainslie et al., 2023). We also train the 2B and 9B models with knowledge distillation (Hinton et al., 2015) instead of next token prediction. The resulting models deliver the best performance for their size, and even offer competitive alternatives to models that are 2-3× bigger. We release all our models to the community.

Die zweite Gemma-Generation (9B & 27B, seit 31.07.24 auch 2B) führte auch zum grandiosen Gemma-2-9b-it-SimPo!

Quelle: https://storage.googleapis.com/deepmind-media/gemma/gemma-2-report.pdf
https://ai.google.dev/gemma/docs/releases?hl=de

Gemma 3

Google DeepMind *2025-03-12*

Gemma 3 Technical Report

Gemma Team, Google DeepMind[1]

We introduce Gemma 3, a multimodal addition to the Gemma family of lightweight open models, ranging in scale from 1 to 27 billion parameters. This version introduces vision understanding abilities, a wider coverage of languages and longer context – at least 128K tokens. We also change the architecture of the model to reduce the KV-cache memory that tends to explode with long context. This is achieved by increasing the ratio of local to global attention layers, and keeping the span on local attention short. The Gemma 3 models are trained with distillation and achieve superior performance to Gemma 2 for both pre-trained and instruction finetuned versions. In particular, our novel post-training recipe significantly improves the math, chat, instruction-following and multilingual abilities, making Gemma3-4B-IT competitive with Gemma2-27B-IT and Gemma3-27B-IT comparable to Gemini-1.5-Pro across benchmarks. We release all our models to the community.

Die dritte Gemma-Generation (1B, 4B, 12B und 27B, später 270M) wurde ab 26.06. ergänzt um Gemma 3n (E2B & E4B)

Quelle: https://storage.googleapis.com/deepmind-media/gemma/Gemma3Report.pdf

Gemma 4

Gemma 4 models are multimodal, handling text and image input (with audio supported on E2B, E4B, and 12B models) and generating text output.

Core Capabilities

Gemma 4 models handle a broad range of tasks across text, vision, and audio. Key capabilities include:

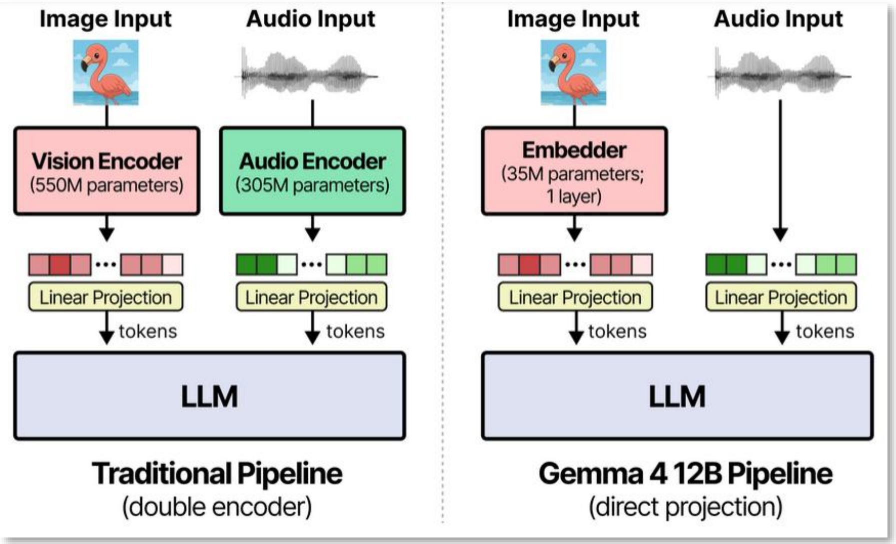
- **Thinking** – Built-in reasoning mode that lets the model think step-by-step before answering.
- **Long Context** – Context windows of up to 128K tokens (E2B/E4B) and 256K tokens (12B/26B A4B/31B).
- **Image Understanding** – Object detection, Document/PDF parsing, screen and UI understanding, chart comprehension, OCR (including multilingual), handwriting recognition, and pointing. Images can be processed at variable aspect ratios and resolutions.
- **Video Understanding** – Analyze video by processing sequences of frames.
- **Interleaved Multimodal Input** – Freely mix text and images in any order within a single prompt.
- **Function Calling** – Native support for structured tool use, enabling agentic workflows.
- **Coding** – Code generation, completion, and correction.
- **Multilingual** – Out-of-the-box support for 35+ languages, pre-trained on 140+ languages.
- **Audio** (E2B, E4B, and 12B Unified only) – Automatic speech recognition (ASR) and speech-to-translated-text translation across multiple languages.

Die vierte Gemma-Generation (E2B, E4B, 31B, 26 A4B; seit 03.06. auch 12B) erstmals unter Apache-2.0-Lizenz

Quelle: https://ai.google.dev/gemma/docs/core/model_card_4

Gemma 4 12B

- A multimodal encoder-free architecture: Bypassing heavy multi-stage vision and audio encoders entirely, multimodal data is fed straight into the LLM backbone, reducing multimodal latency.
- Our first medium-sized model with audio input: In the Gemma family, audio inputs were restricted to small, lightweight edge architectures (e.g. E4B). Gemma 4 12B is the first medium-sized model capable of natively ingesting audio.
- Developer-friendly size: Small enough to run locally on dedicated GPU laptops with 16GB VRAM or unified memory. To maximize local inference speeds, we are additionally releasing a dedicated multi-token prediction (MTP) model.
- New MacOS desktop experience: For the first time, we are releasing downloadable macOS desktop applications, letting developers experience fully local spoken and visual interaction directly on consumer-grade devices.



Quelle: https://developers.googleblog.com/gemma-4-12b-the-developer-guide/

Gemma 4 – Dense Models

Dense Models

Property	E2B	E4B	12B Unified	31B Dense
Total Parameters	2.3B effective (5.1B with embeddings)	4.5B effective (8B with embeddings)	11.95B	30.7B
Layers	35	42	48	60
Sliding Window	512 tokens	512 tokens	1024 tokens	1024 tokens
Context Length	128K tokens	128K tokens	256K tokens	256K tokens
Vocabulary Size	262K	262K	262K	262K
Supported Modalities	Text, Image, Audio	Text, Image, Audio	Text, Image, Audio	Text, Image
Vision Encoder Parameters	*~150M*	*~150M*	-	*~550M*
Audio Encoder Parameters	*~300M*	*~300M*	-	No Audio

The "E" in E2B and E4B stands for "effective" parameters. The smaller models incorporate Per-Layer Embeddings (PLE) to maximize parameter efficiency in on-device deployments. Rather than adding more layers or parameters to the model, PLE gives each decoder layer its own small embedding for every token. These embedding tables are large but are only used for quick lookups, which is why the effective parameter count is much smaller than the total.

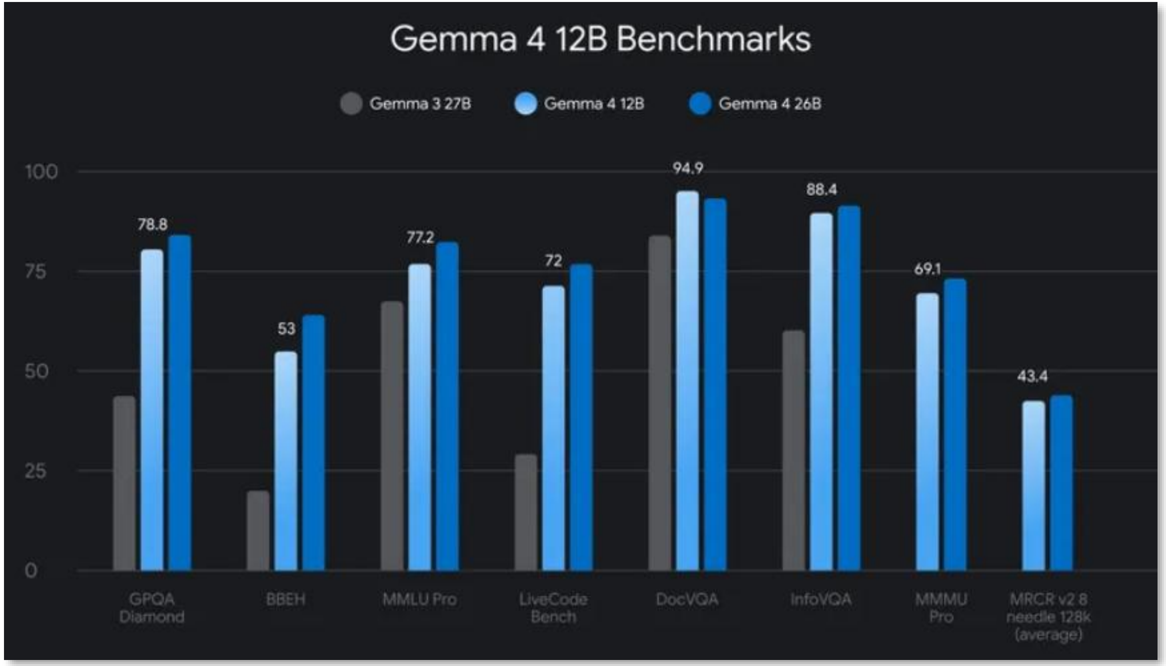
Quelle: https://ai.google.dev/gemma/docs/core/model_card_4

Gemma 4 Benchmarks

Benchmark		Gemma 4 31B IT Thinking	Gemma 4 26B A4B IT Thinking	Gemma 4 E4B IT Thinking	Gemma 4 E2B IT Thinking	Gemma 3 27B IT
Arena AI (text) As of 4/2/26		**1452**	1441	-	-	1365
MMMLU Multilingual Q&A	No tools	**85.2%**	82.6%	69.4%	60.0%	67.6%
MMMU Pro Multimodal reasoning		**76.9%**	73.8%	52.6%	44.2%	49.7%
AIME 2026 Mathematics	No tools	**89.2%**	88.3%	42.5%	37.5%	20.8%
LiveCodeBench v6 Competitive coding problems		**80.0%**	77.1%	52.0%	44.0%	29.1%
GPQA Diamond Scientific knowledge	No tools	**84.3%**	82.3%	58.6%	43.4%	42.4%
τ2-bench Agentic tool use	Retail	**86.4%**	85.5%	57.5%	29.4%	6.6%

Quelle: https://blog.google/innovation-and-ai/technology/developers-tools/gemma-4/

Gemma 4 12B - Benchmarks



Quelle: https://blog.google/innovation-and-ai/technology/developers-tools/introducing-gemma-4-12b/

Gemma 1-4 der Text-Arena

- Mit Blick auf die Kategorie „Text" liegen die evaluierten Gemma-4-Modelle deutlich vor Gemma 3
- Aktuell sind noch nicht alle Modelle evaluiert (und noch keines für die Unterkategorie „German")

Rank	Rank Spread	Model	Score	Votes	Price $/M	Context
41	30 ⊸ 58	gemma-4-31b Google · Apache 2.0	1451 ±8	5.881	$0.14 / $0.40	262.1K
59	42 ⊸ 78	gemma-4-26b-a4b Google · Apache 2.0	1438 ±8	5.804	N/A	N/A
155	147 ⊸ 163	gemma-3-27b-it Google · Gemma	1366 ±4	47.545	$0.08 / $0.16	131.1K
185	164 ⊸ 201	gemma-3-12b-it Google · Gemma	1342 ±10	3.829	$0.05 / $0.15	131.1K
214	198 ⊸ 229	gemma-3n-e4b-it Google · Gemma	1318 ±5	22.601	$0.06 / $0.12	32.8K
236	216 ⊸ 246	gemma-3-4b-it Google · Gemma	1303 ±9	4.171	$0.05 / $0.10	131.1K
243	238 ⊸ 251	gemma-2-27b-it Google · Gemma license	1289 ±3	75.754	$0.65 / $0.65	8.2K
252	240 ⊸ 260	gemma-2-9b-it-simpo Princeton · MIT	1280 ±7	10.072	$0.03 / $0.09	8.2K
262	256 ⊸ 268	gemma-2-9b-it Google · Gemma license	1266 ±4	54.611	$0.03 / $0.09	8.2K
296	292 ⊸ 301	gemma-2-2b-it Google · Gemma license	1200 ±4	46.616	N/A	N/A
308	300 ⊸ 317	gemma-1.1-7b-it Google · Gemma license	1181 ±6	23.893	$0.03 / $0.09	8.2K
335	321 ⊸ 344	gemma-7b-it Google · Gemma license	1136 ±9	8.925	$0.05 / $0.08	8.2K
344	337 ⊸ 349	gemma-1.1-2b-it Google · Gemma license	1115 ±8	10.854	N/A	N/A
350	345 ⊸ 353	gemma-2b-it Google · Gemma license	1092 ±12	4.780	$0.10 / $0.10	N/A

Quelle: https://arena.ai/leaderboard/text/?q=gemma

Gemma 1-3 und die deutsche Sprache

Rank	Rank Spread	Model	Score	Votes	Price $/M	Context
120	48 ⊸ 171	gemma-3-12b-it Google · Gemma	1371 ±43	166	$0.05 / $0.15	131.1K
121	78 ⊸ 150	gemma-3-27b-it Google · Gemma	1371 ±18	1.164	$0.08 / $0.16	131.1K
157	126 ⊸ 199	gemma-3n-e4b-it Google · Gemma	1316 ±22	692	$0.06 / $0.12	32.8K
185	131 ⊸ 225	gemma-3-4b-it Google · Gemma	1289 ±42	162	$0.05 / $0.10	131.1K
198	151 ⊸ 230	gemma-2-9b-it-simpo Princeton · MIT	1269 ±31	318	$0.03 / $0.09	8.2K
199	170 ⊸ 222	gemma-2-27b-it Google · Gemma license	1269 ±14	2.037	$0.65 / $0.65	8.2K
217	186 ⊸ 232	gemma-2-9b-it Google · Gemma license	1245 ±16	1.580	$0.03 / $0.09	8.2K
241	235 ⊸ 254	gemma-2-2b-it Google · Gemma license	1155 ±19	1.158	N/A	N/A
243	233 ⊸ 257	gemma-1.1-7b-it Google · Gemma license	1155 ±22	827	$0.03 / $0.09	8.2K
267	254 ⊸ 270	gemma-1.1-2b-it Google · Gemma license	1048 ±35	365	N/A	N/A

Zum Vergleich:
Mixtral-8x22b-instruct-v0.1
lag bei 1217

Quelle: https://arena.ai/leaderboard/text/german?q=gemma

Diskussion

- Fragen?
- Anregungen?
- Erfahrungen?